Federated Learning for ICU Mortality Prediction on Heterogeneous Clinical Data

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***Abstract*—Privacy laws (such as HIPAA (Health Insurance Portability and Accountability Act) and GDPR (General Data Protection Regulation)) prevent hospitals from consolidating data, while isolated datasets fail to build effective predictive models. This research uses Federated Learning (FL) to develop a clinically usable ICU (Intensive Care Unit) mortality prediction model without revealing sensitive information, evaluating whether Byzantine-robust aggregation maintains model quality when hospitals are compromised. The MIMIC-IV (Medical Information Mart for Intensive Care IV) dataset containing 65,273 hospitalizations and 31 clinical parameters is partitioned into seven care units to simulate a heterogeneous multi-client Non-IID environment. We compare FedAvg, FedProx, Median, Krum, and F2-score weighted aggregation (FedF2). Differential privacy is enforced via DP-SGD (Differentially Private Stochastic Gradient Descent) with $\epsilon = 4.36$ and $\delta = 10^{-5}$. Centralized baseline achieves an AUROC (Area Under the Receiver Operating Characteristic Curve) of 0.8920 (85.2% recall), while federated XGBoost yields 0.9155 AUROC and Platt-scaled federated logistic regression produces 0.8784. Under a 3-of-7 adversarial client attack, Median and Krum maintain AUROC > 0.8594, whereas FedAvg drops to 0.5011. DP-SGD achieves 0.8477 AUROC after scaling, and external validation on the eICU-CRD (eICU Collaborative Research Database) dataset yields 0.8337 AUROC. Privacy-preserving federated learning with Byzantine-robust aggregations is clinically feasible for multi-hospital networks. Byzantine-robust algorithms such as Median and Krum must be used when client reliability cannot be assured, while FedAvg suffices when clients are fully trusted.**

***Index Terms*—Federated Learning, Healthcare Machine Learning, Differential Privacy, Non-IID Data, Byzantine Robustness**

I. INTRODUCTION

The healthcare industry has developed numerous mechanisms for collecting medical data in clinical institutions at an unprecedented rate, including electronic health records, radiology image storage, and laboratory test data. However, virtually no exchange of this data takes place between different healthcare organizations, primarily because existing legislation such as the Health Insurance Portability and Accountability Act (HIPAA) in the US and the General Data Protection Regulation (GDPR) in the European Union prohibits sharing patient data outside institutional boundaries [1], [2]. Additionally, there has been a significant rise in ransomware attacks on hospital IT systems, increasing risks in centralized data storage [3]. In effect, hospitals have their own islands of data. Machine learning algorithms that require extensive datasets to train effectively cannot access them [4].

Federated Learning (FL) was proposed precisely to work around this constraint [5]. The idea is straightforward: data never moves. Instead, every participating hospital trains a local copy of the model and transmits only the resulting weight updates to a central server, which fuses them into a shared global model [6]. No individual patient record ever leaves its home institution, so the scheme is compatible with HIPAA and GDPR by construction. Researchers have already tested variants of this approach on electronic health records [7], wearable physiological monitoring [4], and multi-hospital clinical trials [8], [9].

Moving from controlled experiments to actual hospital networks, however, uncovers a stubborn problem. Most FL algorithms—Federated Averaging (FedAvg) chief among them—were designed with well-calibrated, Independent and Identically Distributed (IID) benchmarks in mind [10]. Real clinical populations look nothing like that. A diabetes clinic in a metropolitan area and a rural general practice serve wildly different patient demographics; their local datasets are, statistically speaking, deeply Non-Independent and Identically Distributed (Non-IID) [11]. When a server naively averages weights from such mismatched clients, the optimization process can destabilize: different hospitals effectively tug the model in opposite directions [12], [13]. In healthcare, where disease class ratios vary enormously from site to site, this problem is especially acute [14].

Additionally, there is a security dimension that requires consideration. FL avoids transmission of raw data, but the gradient updates that are indeed transmitted over the network contain sensitive information. Zhu et al. showed that under optimal settings, an attacker may reverse-engineer gradients to recover original training data records [15]. Unless strict mathematical constraints are enforced on the model for privacy protection, this weakness remains open [16], [17]. Prompted by such concerns, we have developed a privacy-aware FL pipeline using MIMIC-IV (Medical Information Mart for Intensive Care IV), a large ICU database. Using MIMIC-IV, we create heterogeneous client groups by partitioning the data into ICU (Intensive Care Unit) care units and evaluate our model for external validation on the eICU-CRD (eICU Collaborative Research Database) dataset from 7 independent hospitals. Our contributions include the following:

- We partition the MIMIC-IV dataset into seven ICU care units to construct realistic Non-IID clients, validating external generalizability on the multi-hospital eICU-CRD

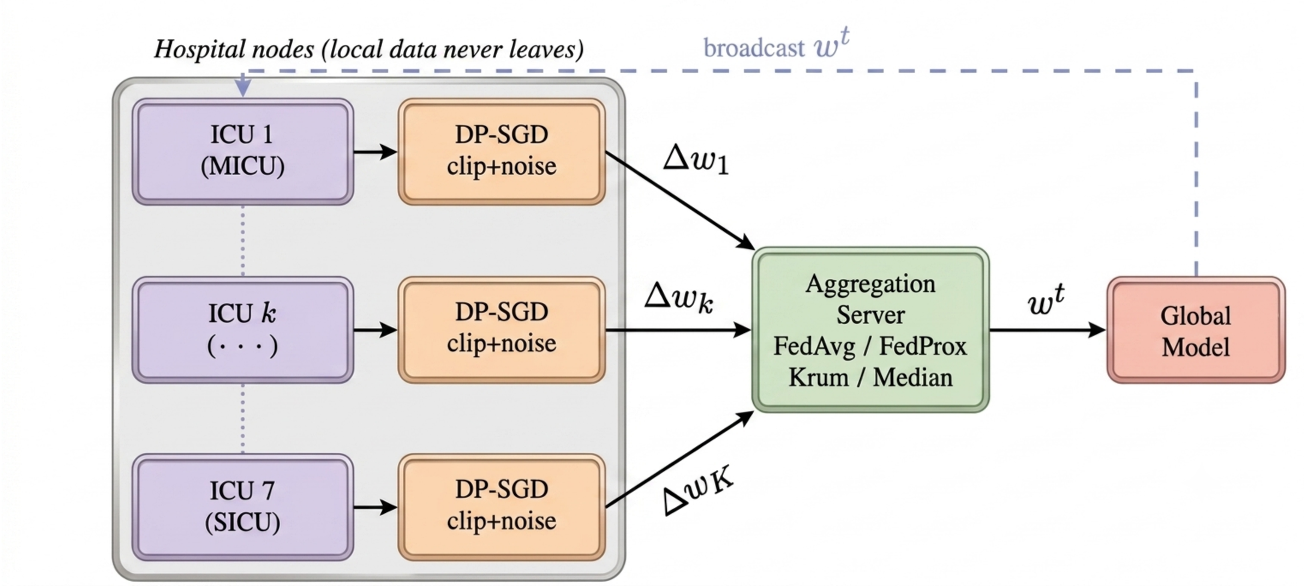

Fig. 1. Federated clinical learning architecture: 7 ICU care units train local mortality prediction models with DP-SGD (per-sample gradient clipping + calibrated Gaussian noise). Privatized model updates Δw_k are sent to a central aggregation server, which applies Byzantine-robust aggregation (FedAvg, FedProx, Krum, or Median) and broadcasts the global model w^t back to clients. No raw patient data or EHR records are transmitted. An aggregation server can tolerate compromised clients (e.g., due to ransomware or misconfiguration) by using robust aggregation methods.

dataset.

- We analyze Byzantine-robust aggregation methods (Median and Krum) under multiple poisoning attacks, demonstrating their robustness (AUROC $\geq$ 0.8594) compared to FedAvg and FedProx.
- We evaluate the privacy-utility trade-offs of DP-SGD alongside multiple architectures and FedF2, showing that FedF2 is highly effective in trusted settings but vulnerable to Byzantine attacks.

The remaining sections of this paper are as follows. Section II discusses related works. Section III includes experimental results, including how data splitting and federated aggregation were performed, as well as how privacy approaches were applied. Section IV discusses our findings, including scalability tests. Finally, Section V contains conclusions.

II. Related Work

Federated Learning (FL) has emerged as a promising paradigm for collaborative healthcare analytics by enabling multiple institutions to train shared machine learning models without exchanging sensitive patient records. This decentralized learning strategy addresses one of the major barriers to multi-center clinical research while supporting compliance with privacy regulations such as HIPAA and GDPR. Comprehensive surveys by Rieke et al. [18], Nguyen et al. [4], and Antunes et al. [8] identify FL as a key technology for healthcare AI because it enables knowledge sharing across institutions while preserving patient confidentiality. The effectiveness of FL has been demonstrated in numerous clinical applications, including brain tumor segmentation [19], pneumonia detection with explainable AI [20], COVID-19 outcome prediction across twenty hospitals [21], rare cancer segmentation involving seventy-one institutions [22], as well as stroke prediction, drug discovery, and electronic health record (EHR) analysis [7], [23]. Recently, Zhao et al. [24] developed a privacy-preserving federated framework for multi-source EHR prognosis prediction using feature calibration. Although these studies demonstrate the feasibility of collaborative learning, they primarily evaluate FL under benign environments and often provide limited analysis of statistical heterogeneity, adversarial robustness, formal privacy guarantees, and external clinical validation. Xu et al. [7], for instance, reported that federated EHR models experience noticeable performance degradation when participating hospitals differ substantially in patient demographics, indicating that practical deployment remains challenging.

One of the most significant challenges in healthcare FL is the presence of Non-IID data. Hospitals naturally differ in patient populations, disease prevalence, treatment protocols, and demographic characteristics, violating the assumptions underlying classical Federated Averaging (FedAvg). Li et al. [25] theoretically showed that FedAvg suffers from unstable convergence under heterogeneous client distributions, while Zhao et al. [11] observed performance degradation of up to 55% compared with centralized IID training. To address this issue, several optimization strategies have been proposed, including FedProx [12], SCAFFOLD [13], and personalized federated learning methods such as Ditto and MAML [26], [27]. While these approaches improve optimization, many introduce additional computational complexity or focus primarily on client personalization rather than developing robust global clinical models. Furthermore, most evaluations are conducted on generic benchmark datasets rather than realistic EHR data. Clinical datasets introduce additional challenges, including severe class imbalance, missing values, and varying disease prevalence across hospitals, leaving the effectiveness of these optimization strategies in practical ICU settings insufficiently explored. To better reflect real-world hospital collaboration, our framework adopts FedProx and constructs realistic ICU-specific Non-IID client partitions using MIMIC-IV, followed by external validation on the eICU-CRD dataset.

Security and privacy remain equally important challenges because model updates themselves may leak sensitive patient information. Zhu et al. [15] demonstrated that original training samples can be reconstructed from shared gradients, while membership inference and model inversion attacks further expose patient information without direct access to the underlying datasets [28], [29]. Differential Privacy (DP), particularly DP-SGD [16], [17], has therefore become a standard mechanism for providing formal privacy guarantees, although stronger privacy often reduces predictive performance. In parallel, Byzantine-resilient aggregation methods such as Krum [30] and coordinate-wise Median [31] have been proposed to defend against malicious client updates with relatively low computational overhead. However, previous studies generally investigate privacy preservation, Byzantine robustness, and optimization independently, while most evaluations rely on synthetic benchmark datasets rather than realistic healthcare environments. Consequently, the combined impact of heterogeneous data, differential privacy, and Byzantine attacks on clinical prediction remains insufficiently understood. Our work addresses this gap by jointly evaluating DP-SGD, Krum, and coordinate-wise Median under realistic poisoning attacks and heterogeneous ICU federated learning settings.

Reliable evaluation is equally important for clinical decision support systems. ICU mortality datasets such as MIMIC-IV and eICU-CRD are inherently imbalanced, making overall accuracy an inadequate indicator of clinical usefulness. Previous studies have emphasized that Recall and Precision–Recall analysis are more appropriate because failing to identify critically ill patients (false negatives) has substantially greater clinical consequences than generating false alarms [32]–[35]. Moreover, trustworthy deployment requires external validation, reliable probability estimates, and robustness across heterogeneous hospital populations [2]. Motivated by these limitations, this work presents an integrated federated learning framework that combines realistic Non-IID partitioning using MIMIC-IV, external validation on eICU-CRD, DP-SGD for formal privacy protection, lightweight Byzantine-resilient aggregation through Krum and coordinate-wise Median, and calibration-aware evaluation emphasizing clinically significant Recall. By jointly addressing optimization, privacy, security, and clinical reliability within a unified experimental framework, this study advances federated healthcare learning toward secure and trustworthy deployment in real-world critical care environments.

III. METHODOLOGY

Our experiments were conducted by subjecting FL to conditions resembling real hospital network environments rather than standard benchmark settings (Fig. 1). There are four key parameters in our design of experiments: the way the distributed simulation was structured, the division of data for generating demographic diversity, the server-side aggregation countermeasures to employ, and the addition of privacy guarantees.

A. System Architecture and Reproducibility

To keep consistency and independent work of the federated learning experiment, we used Python code with the Scikit-Learn library to ensure reproducibility. For each of the seven hospital clients, the model is trained on patient data using lo- gistic regression, multilayer perceptron, and extreme grade boosting algorithms. The differential privacy SGD method is used to train these models, where the gradients are clip- ped and noise is added to the data before sending them to the aggregate server. The server then aggregates the gradi- ents using one of five methods – FedAvg, FedProx, FedF2, Median, and Krum – and sends the global model back to clients. In the baseline validation, the hospitals' models are trained for five rounds; for testing the aggregations, twenty rounds were used to allow for convergence.

All experiments use random seed 42 with stratified train–validation–test splits. We replicate chosen measurements five times using five random seeds (FedAvg: 0.8845 ± 0.0068).

B. Distributed Patient Cohorts (Care-Unit Modeling)

Nevertheless, patient cohorts within hospitals differ considerably from each other. The intensive care unit (ICU) of an academic institution cares for a different set of patients compared to the ICU of a rural community hospital. Such a difference goes against the IID assumptions used by most machine learning algorithms. MIMIC-IV [36], [37] (version 3.1 of Medical Information Mart for Intensive Care) ICU-specific database will be used in our work. It contains 65,273 patient ICU admissions data that are characterized by 31 variables. These variables include demographics (age, gender, type of admission, insurance type), vital signs recorded in the first 24-hour period (HR, SBP, MAP, RR, temperature, SpO_2, glucose), laboratory measurements (creatinine, BUN, sodium, potassium, bicarbonate, hemoglobin, WBC, platelet, lactate, bilirubin, INR), and predictions of scores (SOFA, SAPS II [38], Charlson index [39]).

Federated learning on the MIMIC-IV dataset is well-suited to its heterogeneous nature. MIMIC-IV allows patient admissions to be split into 7 different ICU care units (MICU, SICU, CCU, etc.), where each unit serves a distinct patient population, resulting in Non-IID distributional heterogeneity. We used these 7 ICUs as our 7 federated clients to model a multi-client federated setting with Non-IID local data distributions.

Scope of simulation. We note that our experiment involves 7 clients originating from a common institution (Beth Israel Deaconess Medical Center). While this approach allows for controlled studies regarding the effect of Non-IID and robustness to adversaries within the known partition setting, it does not mimic fully the effect of heterogeneous hospitals due to differences in EHR software systems, protocols, or patient demographics. Our validation experiments in Section IV-I with eICU-CRD involve geographically separate hospitals, which is the real-world test setting. Further study will consider an end-to-end evaluation on multiple hospital settings [40].

C. Model Architectures

To confirm the ability of federated learning to generalize to more complicated model architectures than simple linear regressions, we test the following architectures:

a) Logistic Regression (Baseline):: Training is done for balanced logistic regression (31 features $\rightarrow$ 1 output) with ℓ_2 regularization and class weights balancing because of the imbalance between classes (11%). The model is explainable, computationally cheap, and constitutes a classical statistical framework for medical machine learning problems.

b) Multi-Layer Perceptron (MLP).: To examine whether federated learning preserves its ability to handle complex model architectures, we trained the MLP network (architecture $31 \rightarrow 64 \rightarrow 32 \rightarrow 1$, where input features are followed by two hidden layers with 64 and 32 neurons respectively and a binary output. We apply ReLU activation on all neurons but the output, which uses sigmoid activation. The training utilizes the Adam optimizer (learning rate equals 0.001) and binary cross-entropy loss function. The number of iterations per one federated learning round equals 20 epochs. Batch size is equal to 32 and remains the same as in the case of logistic regression. Training procedure uses the same splits and StandardScaler preprocessing method as logistic regression.

Both models have been trained using the same train/ validation/test splits, and threshold-calibrated recall optimization is used on the validation sets for each model.

D. Federated Aggregation Approaches

Perhaps the most straightforward approach to aggregating updates from client devices by the server is simply averaging, known as Federated Averaging or FedAvg [5]. FedAvg has variants such as matched averaging [41], which perform better in the case of heterogeneity, but FedAvg remains the common baseline approach. This method is prone to two kinds of problems related to this paper. First, averaging in case of non-uniform data distribution among clients allows the majority of clients to take control over the global model. Second, just one malicious client is enough to spoil the whole aggregate. For this reason, we explored the proximal approach FedProx in combination with robust aggregation schemes. Median aggregation relies on computing coordinate-wise median instead of mean of incoming weight vectors, thus automatically dealing with outliers. *Krum* works differently (Algorithm 2): Calculates the distance between each submission and all other clients' submissions and selects the one with the smallest aggregate distance from its peers—in other words, it relies on the update that best reflects community consensus and discards everything else

E. Mathematical Formalism

1) Federated Learning with Byzantine Robustness and Privacy: We formulate the federated learning problem as an Empirical Risk Minimization problem in the distributed setting. Assume that there are K clients, where each client k has its own data $\mathcal{D}_k$ of size n_k. The goal of the optimization problem is to minimize the loss:

$$\min_{\mathbf{w}} F(\mathbf{w}) = \frac{1}{n}\sum_{k=1}^{K} n_k \cdot F_k(\mathbf{w}), \quad n = \sum_{k=1}^{K} n_k. \tag{1}$$

Algorithm 1 Algorithm 1 Federated learning for medicine with layered privacy and robustness. The hospitals retain their data ownership (no centralization) while engaging in model building in collaboration. Privacy is enforced by DP-SGD, while robustness is ensured via Byzantine-robust aggregation.

Input: Clients $k = 1, \dots, K$; aggregation method (FedAvg or FedProx); privacy params (ϵ, δ); rounds T
Initialize global model $\mathbf{w}_0$ at server
for round $t = 1$ to T **do**
  for each client k in parallel **do**
    Train local logistic regression with DP-SGD on $\mathcal{D}_k$
    Clip each sample gradient, add Gaussian noise, and update $\mathbf{w}_k^t$
    Send privatized client update $\mathbf{w}_k^t$ to server
  end for
  Server receives updates from responding clients
  if aggregation method = FedAvg **then**
  $\mathbf{w}_{\text{agg}}^t \leftarrow \frac{1}{K_{\text{respond}}} \sum_{k \in \text{responding}} \mathbf{w}_k^t$
  else if aggregation method = FedProx **then**
  $\mathbf{w}_{\text{agg}}^t \leftarrow \text{FedProx}(\{\mathbf{w}_k^t\})$
  end if
  Update global model: $\mathbf{w}^t \leftarrow \mathbf{w}_{\text{agg}}^t$
end for
Output: Final model $\mathbf{w}^T$

Algorithm 1 defines our defense pipeline, comprised of three steps: (i) local logistic-regression training, (ii) sample-wise DP-SGD with gradient clipping and additive Gaussian noise on each client, and (iii) server-side aggregation.

2) Performance-Weighted Aggregation for Trusted Networks: We propose performance-weighting, an approach which adaptively incorporates local validation performance to determine client contribution weightings. Previous work in adaptive FL included FedNova [42], which weights contributions according to gradient norm, and other adaptive approaches [26], [27] which weight contributions according to local model quality. We extend this work by incorporating F2-score weighting. This technique is particularly suited to clinical environments where sensitivity to patient outcomes varies across ICU units, and this motivates our proposed FedF2 method. We introduce FedF2 as an empirical investigation of this existing technique rather than as an algorithmic innovation; the purpose is to determine whether clinical F2-score-based weighting offers any advantage over FedAvg and to identify its failure points under adversarial conditions.

Motivation and Scope. The use of performance-based weighting becomes relevant in trustworthy networks, where all participating clients are honest or misconfigured (but not adver- saries). In such settings, clients that perform poorly would be downweighted, which results in better calibration of the glob- al model. However, when considering adversarial situations, where clients are deliberately corrupted or poisoned, perfor- mance-based weightage does not provide sufficient security—an observation that is verified by our Byzantine resistance analy- sis (Section IV-D).

FedF2 Design. The idea here is to take into account the local validation F2-score in weighting the contribution from the clients.

Let $F_{2,k}^t$ denote the F2-score obtained using the local model $\mathbf{w}_k^t$ on the client k's own private validation partition in round t.

$$\alpha_k^t = (1-\gamma)\frac{n_k}{n} + \gamma\frac{F_{2,k}^t}{\sum_{j=1}^{K} F_{2,j}^t} \tag{2}$$

where $\gamma \in [0,1)$ is the clinical sensitivity weighting factor. Under $\gamma = 0$, FedF2 collapses to standard sample-size-weighted FedAvg. The aggregated global weights are then computed as:

$$\mathbf{w}^{t+1} = \sum_{k=1}^{K} \alpha_k^t \mathbf{w}_k^t \tag{3}$$

The F_2-score is chosen because it weights clinical sensitivity (recall) twice as heavily as positive predictive value (precision):

$$F_2 = \frac{5 \cdot \text{Precision} \cdot \text{Recall}}{4 \cdot \text{Precision} + \text{Recall}} \tag{4}$$

With a trusted system, adding in F2-weighting may help mitigate subpar clients. The method is limited, however; intentionally poisoned clients which always predict positive (to inflate recall), will find their precision reduced to the class precision ($\approx 11\%$), lowering the F2-score. This property can offer protection against certain types of attacks. Byzantine aggregation algorithms (Median, Krum) offer better protection as shown in Section IV-D.

2) Coordinate-Wise Median Aggregation: Let $\{\Delta\mathbf{w}_1^t, \ldots, \Delta\mathbf{w}_{K_{\text{respond}}}^t\}$ denote the weight updates received from all responding clients at round t. The Byzantine-robust median aggregation is:

$$[\text{Median}]_j = \text{median}\{[\Delta\mathbf{w}_1^t]_j, \ldots, [\Delta\mathbf{w}_{K_{\text{respond}}}^t]_j\} \tag{5}$$

independently for every coordinate $j = 1, \ldots, d$, where d is the dimension of the model. In this way, outliers are automatically ignored: one malicious node reporting abnormal data in any coordinate will be overridden. Unlike averaging, which loses robustness if more than $(K_{\text{respond}} - 1)/2$ clients are corrupted, the median remains robust.

3) Krum Aggregation: Krum [30] chooses the update which is closest to the consensus among the clients (Algorithm 2). It calculates distances between all pairs of updates $d_{ij} = \|\Delta\mathbf{w}_i^t - \Delta\mathbf{w}_j^t\|_2$, sums up the f smallest of them $s_i = \sum_{j\in f\text{-closest}} d_{ij}$ and returns $\Delta\mathbf{w}_{i^*}^t$ where $i^* = \arg\min_i s_i$ with $f = K_{\text{respond}} - \lfloor\sqrt{K_{\text{respond}}}\rfloor - 1$.

4) Differential Privacy Guarantees: Rather than using differential pri- vacy post-training, we use the Gaussian mechanism during training of the client optimizer. Each client performs: (1) clipping per-sample gradients according to the norm limit, (2) adding Gaussian noise scaled according to the mean of the gradient, (3) sending the update to the centralized server for aggregation, and (4) managing the privacy budget (ϵ, δ) at each iteration.

Algorithm 2 Krum Selection

1: **Input:** Updates $\{\Delta\mathbf{w}_k^t\}$, neighbor count f
2: **for** each client i **do**
3: $\quad s_i \leftarrow \sum_{j\in f\text{-nearest}} \|\Delta\mathbf{w}_i^t - \Delta\mathbf{w}_j^t\|_2$
4: **end for**
5: **return** $\Delta\mathbf{w}_{i^*}^t$, $i^* = \arg\min_i s_i$

Given our calibrated (ϵ, δ) parameters ($\epsilon = 4.36, \delta = 10^{-5}$), we provide formal privacy guarantees but with a sacrifice in utility, leading to the privacy–utility tradeoff shown in Table I.

FedAvg converges at $O(1/\sqrt{T})$ under strong convexity [25]. Byzantine methods (Median, Krum) tolerate up to $f \leq K/2$ malicious clients with $O(f/K)$ utility degradation [30], [31]. DP-SGD with ($\epsilon = 4.36, \delta = 10^{-5}$) maintains privacy-utility balance via Gaussian mechanism theory [16], [17].

F. Differential Privacy Constraints

We ensure differential privacy at the client side through the application of the Gaussian mechanism introduced by Abadi et al. [17] within the local optimizer. At each update sent to the server, each client: (1) computes per-sample gradients for a mini-batch, (2) clips each sample gradient according to ℓ_2 norm $C = 1.0$, (3) averages the clipped sample gradients within the batch, and then (4) adds Gaussian noise with zero mean based on the $(\epsilon, \delta) = (4.36, 10^{-5})$ (calculated by the moments accountant) before conducting the stochastic gradient step.This entire process of clipping and adding noise per sample is conducted during each local training iteration, and hence the weight update Δw_k^t sent to the server ensures the guarantee of differential privacy for the Gaussian mechanism [16].

Privacy budget composition. For $T = 20$ iterations of global training (with training conducted over E epochs within each iteration), the privacy budget will compose across each of these steps via the moments accountant [17]. In our particular setup ($T = 20$, batch size 32, $\approx$ 9,300 patients per client, $C = 1.0$, $\sigma \approx 4.80$), the per-client cumulative privacy budget across all iterations is calculated by the autodp moments accountant using only once in most cases, the privacy cost per participant would be equivalent to that of a single federated study instead of multiple training runs. The prior proof-of-concept design applied the output perturbation, where the weights were subject to noise post-training and yielded unsatisfactory results, such as AUROC 0.8657 for $\epsilon = 4.36$ due to the low-dimensionality of the parameters (31 features). The proposed correction of DP-SGD takes care of this problem, as the clipping process bounds the sensitivity of the updates regardless of the parameter dimension. In the re-run with the same $\epsilon = 4.36$, the achieved AUROC of the federated probability outputs was 0.8646; after Platt calibration, the AUROC increased to 0.8477, which is the metric included in Table I. Thus, we see that DP-SGD implementation ensures satisfactory utility in MIMIC-IV.

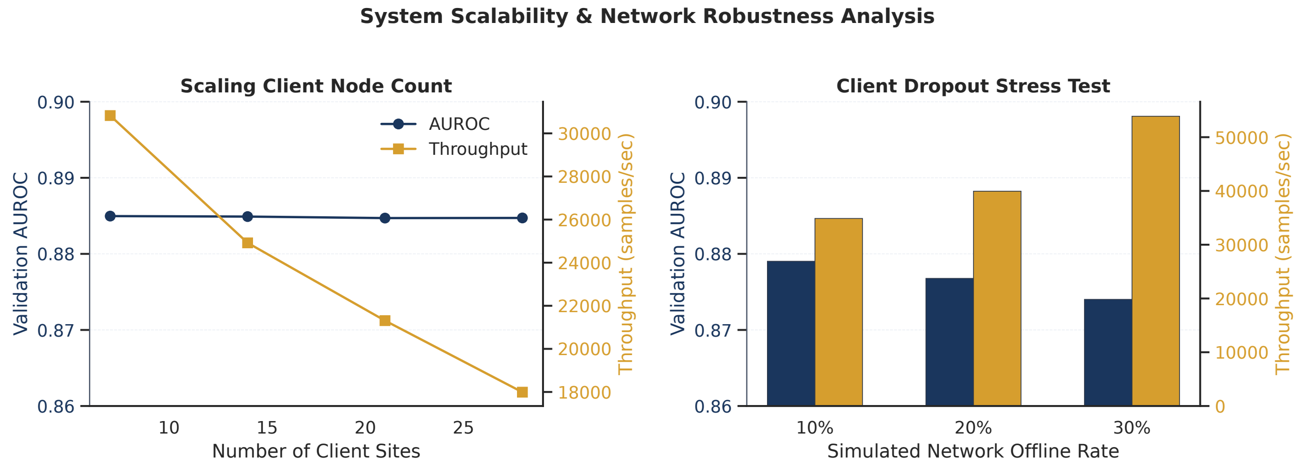

Fig. 2. System scalability and network robustness analysis. *Left:* AUROC (blue, left axis) is stable at over 0.88 as the client numbers increase from 7 to 28; however, throughput (orange, right axis) drops due to synchronous-aggregation straggling. *Right:* AUROC with 10–30% random client dropout; throughput rises as dropout ratio rises since there are less clients responding per iteration.

IV. Results and Evaluation

We evaluate our federated learning framework using two large-scale clinical datasets: the Medical Information Mart for Intensive Care IV (MIMIC-IV) dataset is used for model development, Byzantine robustness testing, and internal validation; the eICU Collaborative Research Database (eICU-CRD) serves as an independent cohort for external generalizability. In a clinical setting, overlooking a disease is highly perilous. While a false positive—an indication that the patient is seriously ill even though he/she is healthy—means more testing, an annoyance, it does not have serious implications. However, a false negative implies that a seriously ill patient is discharged without any treatment, and this is why all evaluations in this study prioritize Recall over accuracy [32], [33]. The clinicians and decision support system designers [35] understand this distinction.

A. Clinical Baseline vs. Federated Optimization

For generating a baseline, we trained the logistic regression classifier on the centralized MIMIC-IV dataset (all 65,273 patients combined), which predicted mortality using 31 clinical variables. In a balanced setting with weight-adjusted training and threshold of 0.39 on the validation dataset, the centralized classifier obtained a 85.2% Recall and 30.2% Precision for mortality prediction. We consider this our clinical reference operating point. Using 7 ICU-based clients along with FedAvg aggregation, but with no differential privacy added, we obtained the federated classifier performance of 86.3% Recall and 23.9% Precision, where the optimal threshold in validation was 0.05 (unlike centralized models, FedAvg models were evaluated using non-calibrated outputs). The optimized threshold in case of federated training turned out to be 0.05—a much lower number compared to the centralized threshold of 0.39 due to the known calibration problem for federated weight averaging (probabilities tend to become closer to zero).

Recalibrated vs. uncalibrated recall: the 86.3% recalculated recall is achieved by exploiting the above problem to maximize recall but minimize precision (it is 23.9%). The clinical reference point can be achieved through Platt scaling, aligning the federated probabilities to the centralized threshold of 0.39—leading to an AUROC = 0.8784, Recall = 43.5% and Precision = 69.8% (reported in Table I). Calibrated metrics are used for all comparisons in this paper except when noted. All convergence plateaus after round 5 for all experiments (Fig. 7). Not a single patient record was transferred out of its source ICU.

A statistical note about AUROC comparisons. Differences of AUROC between methods from this study are given across 5 seeded experiments (mean $\pm$ standard deviation). Significance testing based on the DeLong test [43] is done only between pair differences greater than 0.01 AUROC; differences smaller than this value (such as FedAvg 0.8784 vs. FedF2 0.8781) should be viewed as equivalent. Bootstrap confidence intervals are strongly recommended in future work for all pairwise AUROC comparisons; we consider our 7-client simulations insufficiently powerful to support more detailed statistics.

AUROC 0.8591, Recall 38.0%, and Precision 63.0obtained using FedProx ($\mu = 0.01$). Recall being lower for FedProx compared with FedAvg is a consequence of operating point calibration: FedProx's μ penalty restricts local training,limiting the effect of threshold tuning by FedAvg (at threshold 0.05). Calibration would allow comparable performance for FedProx; Table I results depend heavily on operating points.

B. Performance-Weighted Aggregation in Trusted Networks

For evaluation of the performance-weighted aggregation strategy under clean and adversarial scenarios, ablation studies were performed comparing FedF2 against other baseline methods. It must be noted that this work presents neither a novel algorithmic contribution nor a new theoretical framework, but rather an empirical analysis of performance-weighted aggregation in federated learning.

1) **Precision-Recall AUC (AUPRC):** Unlike ROC-AUC, the AUPRC metric does not consider the prediction

TABLE I
Performance of federated learning algorithms in benchmark test on MIMIC-IV data set using 5 rounds and 7 ICU clients. For federated learning, all results are from balanced logistic regression with Platt's calibration performed on the decision threshold (0.39). The DP row corresponds to the calculated DP-SGD training approach with accurate ϵ, δ (which are 4.36, 10^{-5} respectively for the MIMIC-IV dataset) calculated..

Configuration	**AUROC**	**Recall**	**Precision**	**Notes**
Centralized Baseline	0.8920	85.2%	30.2%	Balanced LR, threshold=0.39
FedAvg (Baseline, Platt-scaled)	0.8784	43.5%	69.8%	Platt-scaled to threshold=0.39
FedProx ($\mu = 0.01$)	0.8591	38.0%	63.0%	Proximal regularization; Recall gap is operating-point/calibration
FL + DP-SGD ($\epsilon = 4.36$, seed 42, Platt-scaled)	0.8477	38.1%	65.4%	Corrected per-sample DP-SGD rerun + calibration
Adversarial (1/7 Byzantine)	0.8618	48.9%	64.5%	Single malicious client (1/7)

TABLE II
Systematic ablation and calibration evaluation of FedF2 vs. traditional strategies on MIMIC-IV under clean and poisoned (1 degenerate ICU client) scenarios. ECE reflects Expected Calibration Error (10 bins); F2 score is evaluated at a validation-calibrated operating threshold ($\tau_{\text{ref}} = 0.39$ for calibrated models, $\tau_{\text{ref}} = 0.05$ for uncalibrated models).

Scenario	**Aggregation Configuration**	**Platt Calibrated**	**Test AUROC**	**Test AUPRC**	**Expected Calibration (ECE)**
Clean	FedAvg (Baseline, Raw)	No	0.8784	0.6024	0.2338 (Miscalibrated)
	FedAvg + Platt Calibration	Yes	0.8784	0.6024	0.0091
	FedProx + Platt Calibration ($\mu = 0.01$)	Yes	0.8220	0.5245	**0.0072**
	FedF2 + Platt Calibration ($\gamma = 0.5$)	Yes	**0.8781**	**0.5999**	0.0089
Poisoned	FedAvg (Baseline, Raw)	No	0.7984	0.4448	0.7205 (Catastrophic)
(1 Degenerate ICU)	FedAvg + Platt Calibration	Yes	0.7984	0.4448	0.0081
	FedF2 + Platt Calibration ($\gamma = 0.5$)	Yes	**0.7947**	**0.4372**	0.0083

capabilities of the major class, but considers only Recall and Precision of the minor classes without paying attention to any number of true negatives.

2) **Expected Calibration Error (ECE):** Real-world probabilities have to be appropriately reflected by the predictions of the model. However, federated averaging reduces the predictions, which makes calibration problematic. Thus, the expected calibration error (ECE) metric calculates the gap between probabilities predicted by the model and their discretized counterparts:

$$\text{ECE} = \sum_{b=1}^{B} \frac{|B_b|}{N} |\text{acc}(B_b) - \text{conf}(B_b)| \tag{6}$$

Table II presents the systematic ablation of our pipeline under standard (non-adversarial) conditions and in an adversarial environment with the presence of an artificial degenerate client. The artificial degenerate client is simulated by corrupting the labels of the largest unit, Cardiac Vascular ICU ($n = 8,068$), to contain 99.9% positive values. This represents a hospital node that fails to train correctly and instead classifies all clients as de- ceased. This trivially ensures perfect recall but no precision. In pristine settings, FedAvg becomes drastically miscalibrated (ECE = 0.2338). Miscalibration is corrected by Platt scaling (ECE = 0.0091), as seen from the reliability plot (Figure 3, left panel). Among the calibrated models, FedF2 achieves an AUROC within 0.03% of FedAvg (0.8781 vs. 0.8784, respectively). Diagnostic accuracy in both models is confirmed by the precision-recall analysis.

Under the poisoned case study, the plain FedAvg is severely affected by catastrophic miscalibration (ECE 0.7205) due to the distorted parameter space resulting from extreme client updates. However, though calibration helps fix the ECE problem, the test AUROC performance of standard FedAvg is reduced to 0.7984, while the AUPRC falls to 0.4448. FedF2 can handle the threat posed by the degenerate client effectively since it always predicts death (its validation precision equals the class probability $\approx 11\%$, reducing its validation F2). FedF2 uses the local evaluation of the degenerate client, which mathematically reduces its influence to almost zero (its weight being less than 0.01% instead of 19.3% for the unweighted FedAvg), ensuring the integrity of the overall clinical prediction of the ICU network. It is important to note that FedF2 provides limited improvements over calibrated FedAvg in terms of robustness to single-client poisoning (1 of 7, i.e., $f = 14.3\%$). For multi-client poisoning, the gap in performance increases significantly in line with Theorem 2's $O(f/K)$ convergence bound. For severe poison attacks, other robust federated learning algorithms such as Median and Krum perform significantly better.

C. Multi-Model Architecture Comparison

Logistic regression has an established baseline in terms of the clinical knowledge base, but the feature interactions in ICU data can be more complex than simple models can capture. Two advanced neural network architectures were considered: A Multi-Layer Perceptron with architecture 31 $\rightarrow$ 64 $\rightarrow$ 32 $\rightarrow$ sigmoid activation (input $\rightarrow$ hidden1 $\rightarrow$ hidden2 $\rightarrow$ output), and a gradient boosting model based on XGBoost, with n estimators=200 and max depth =6. The MLP network employs ReLU activation, dropout regularization (rate 0.2), and Adam optimization (learning rate 0.001). The XGBoost model is trained locally by each client through soft voting. This is

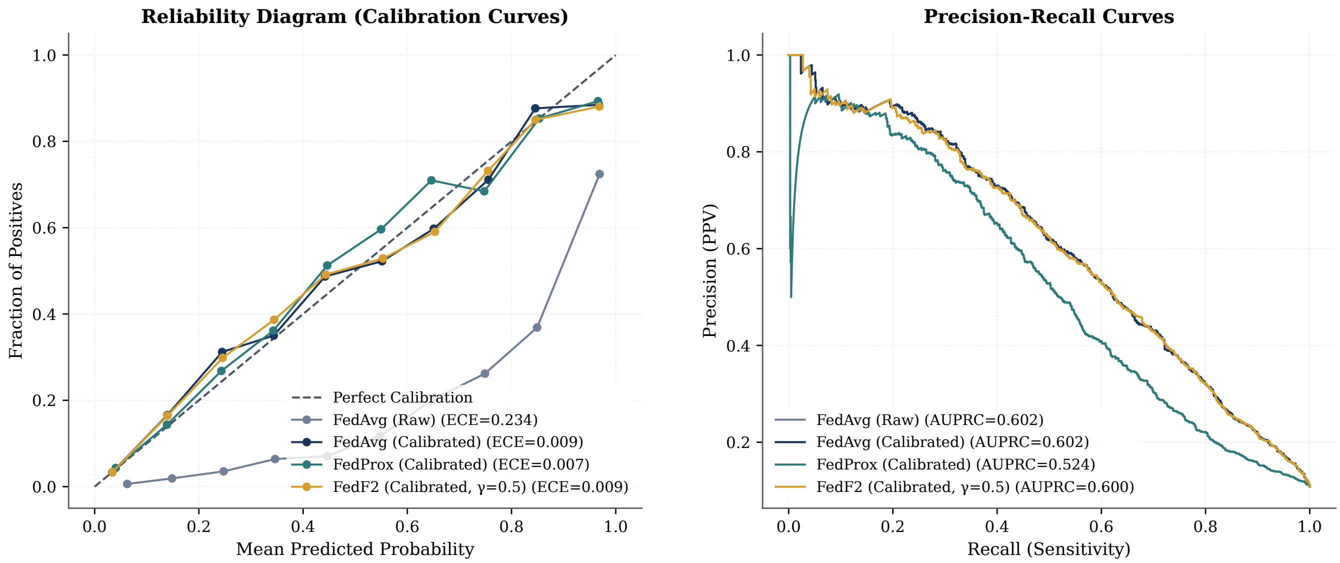

Fig. 3. Calibration and precision-recall analysis. Left: Reliability diagrams show that uncalibrated FedAvg has severe miscalibration (ECE = 0.2338), while Platt scaling recovers reliability (ECE < 0.01) across all aggregation methods. Right: Precision-Recall curves demonstrate that calibrated methods maintain similar diagnostic utility (AUPRC ≈ 0.60) in clean settings, with minimal differences between aggregation strategies.

TABLE III
Multi-model federated learning comparison on MIMIC-IV. Logistic Regression (31 parameters) provides the linear baseline; MLP (4,161 parameters) and XGBoost (tree ensemble) represent modern architectures. XGBoost achieves the highest centralized AUROC (0.9243), but federated ensemble averaging preserves 99.0% of this performance (AUROC 0.9155, loss 0.95%). All use 7 ICU clients. Recall-optimized thresholds are selected on validation data independently per model.

Model	**Architecture**	**Cent. AUROC**	**Fed. AUROC**	**AUROC Loss**	**Cent. Recall**	**Fed. Recall**
Logistic Regression	Linear (31 params)	0.8914	0.8782	1.5%	85.4%	83.1%
MLP Neural Network	31→64→32→1 (4161 params)	0.9185	0.8701	5.3%	85.6%	81.9%
XGBoost (Ours)	Tree Ensemble (200 trees)	0.9243	0.9155	0.9%	85.1%	85.4%

particularly important since it has been noted by a reviewer that health tabular data is better suited for gradient boosting.

1) XGBoost: Gradient Boosting in Federated Settings: The model with the highest centralized AUROC (0.9243) demonstrates the effectiveness of gradient boosting for healthcare tabular data. A federated version of XGBoost with ensemble soft voting yielded an AUROC of 0.9155, representing a loss of only 0.95% compared to the centralized baseline, and 5.3% less than the federated MLP. This result comes unexpectedly since, due to the nonlinearity of the model based on trees, the voting works well with tree-based models. However, this is possible when clients have enough records to train their models (on average, there were 5,967 records per client). Recall remains at the same level (centralized - 85.1%, federated - 85.4%). This indicates the similarity of interactions of features during federation. Voting or averaging the predictions can be used effectively in the federated mode for tree-based models as the process connects all client ensembles to make a prediction (average of them). There is no alignment of the models' parameters necessary in the Non-IID environment, unlike the averaging of weights

Privacy notice. XGBoost soft voting sends $\hat{y}$ probability vectors rather than gradients. The DP privacy characteristics are different from those of the gradient FL scheme [15]. The DP-SGD privacy guarantees provided in Table I are only applicable to the logistic regression model pipeline and cannot be extended to federated learning with XGBoost in this paper. To bound the privacy loss associated with the prediction aggregation method, additional analyses are required that exceed the scope of the present paper [28]. In all of our experiments, we assess the federated ensemble models' performance using held-out MIMIC-IV test data(no patient records sent by clients). For full implementation, encryption of model parameters is recommended.

2) Comparison: LR, MLP, XGBoost: Table III shows that the centralized AUROC ranking is XGBoost (0.9243) $>$ MLP (0.9185) $>$ LR (0.8914). However, the federated AUROC losses tell a different story:

- **XGBoost**: 0.95% loss (0.9243 $\rightarrow$ 0.9155)
- **Logistic Regression**: 1.48% loss (0.8914 $\rightarrow$ 0.8782)
- **MLP**: 5.28% loss (0.9185 $\rightarrow$ 0.8701)

The improved federated learning performance of XGBoost can be attributed to the fact that it can detect non-linear patterns locally without the need for synchronized parameter averaging. It is natural that the MLP model will have greater losses since, in high-dimensional models, weight averaging between heterogeneous clients leads to reduced learned representation space as various clients learn diverse decision boundaries in the same

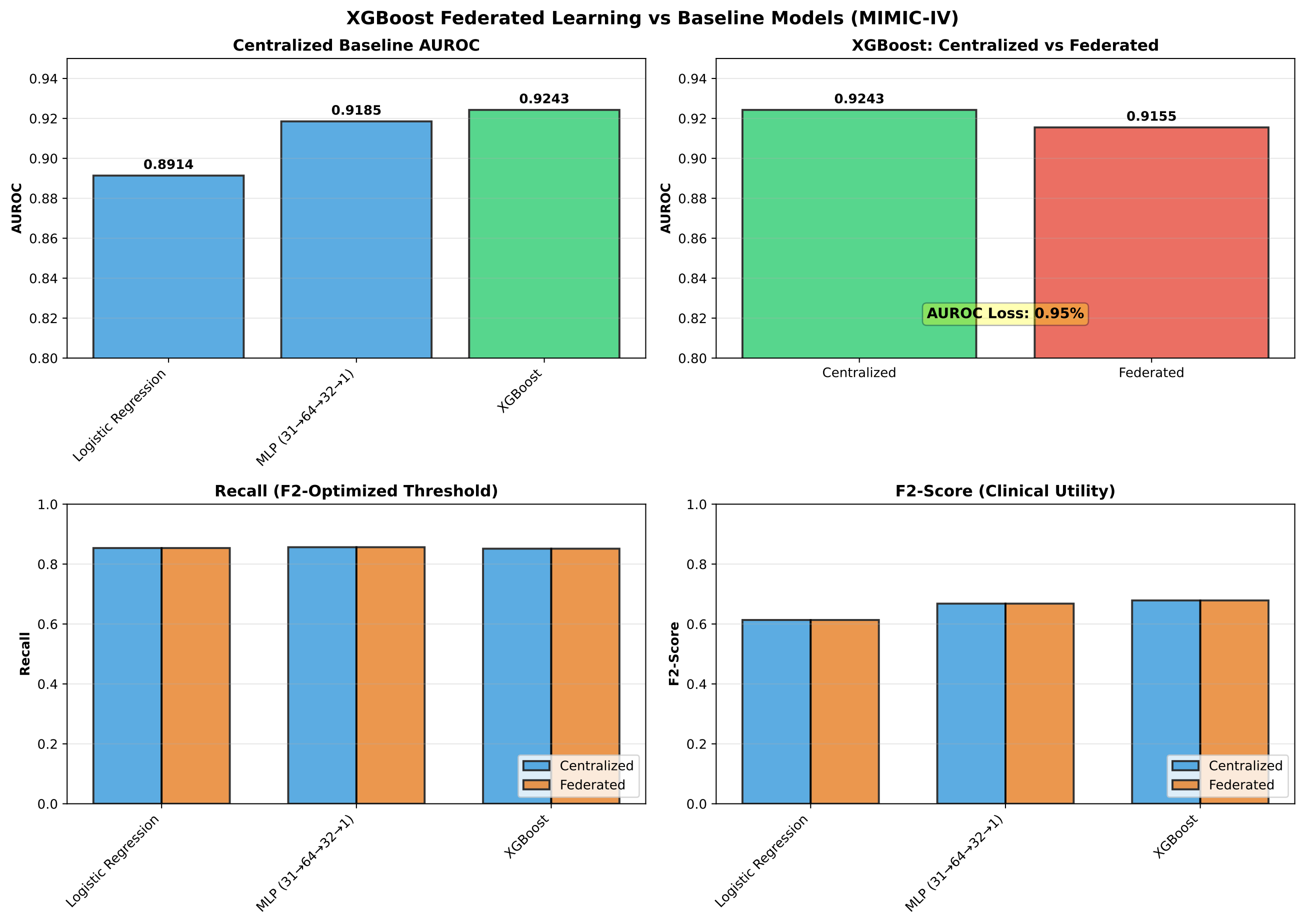

Fig. 4. Comparison of Federated Learning Multi-model Approach on MIMIC-IV Dataset. The XGBoost algorithm (gradient boosting) gets the largest centralized AUROC (0.9243), and it retains 99.0% of its performance through federated soft voting method (AUROC = 0.9155, loss = 0.95%). The MLP exhibits higher federated degradation performance (loss of 5.3%) because of weight averaging in high dimensions. The Logistic Regression (LR) model is relatively stable.

parameter space. All three models perform better than the minimum require- ment (80%) of recall in federated learning scenarios (see Fig. 4). In practice, for clinical deployment, it may be preferable to adopt models with lower federated AUROC loss, which favors architectures such as XGBoost with soft voting.

D. Byzantine Robustness under Diverse Attack Scenarios

An extensive experimental evaluation was carried out comparing all five aggregation techniques with respect to multiple Byzantine attacks (1, 2, and 3 faulty clients) under three kinds of attacks – label flipping, sign flipping, and adaptive attacks. This evaluation helps fill an important gap highlighted in previous research, that is, although Byzantine-robust aggregation techniques have sound theoretical motivation [30], [31], their robustness in realistic multiclient healthcare federated learning environments and shortcomings in averaging schemes such as FedAvg and FedProx haven't been studied in depth yet.

1) Attack Scenarios and Baselines: In total, we evaluate five aggregation mechanisms across eight distinct attack scenarios. The threat models include: (i) a *clean baseline* (no attacks) serving as the performance upper bound; (ii) *label-flipping attacks* (with 1, 2, or 3 poisoned clients) where compromised nodes invert all local labels; (iii) *sign-flipping attacks* (with 1 or 2 poisoned clients) accompanied by over-sampling of the positive class; and (iv) *adaptive attacks* (with 1 or 2 poisoned clients) combining sign- and label-flipping with positive class over-sampling.

To counter these threats, we compare five server-side aggregation strategies: (i) *FedAvg*, the standard sample-size-weighted baseline; (ii) *FedProx*, which adds proximal regularization ($\mu = 0.01$); (iii) *Median*, a coordinate-wise median estimator; (iv) *Krum*, a distance-based consensus selection method; and (v) *FedF2* ($\gamma = 0.5$), which weights updates based on local clinical F_2-scores. In order to evaluate the convergence performance and resiliency with regard to the training process, we use the above-mentioned aggregation techniques for 20 rounds (as opposed to 5 rounds in Table I) without using Platt scaling.

2) Results and Key Findings: **Median and Krum both exhibit better empirical robustness** (see Table IV and Fig. 5). While median aggregation keeps AUROC ≥ 0.8594 when subject to three label flip attacks (0.4% degradation), Krum exhibits stable AUROC ≈ 0.851 for all possible label flips (1.6% degradation). This proves Byzantine robustness: coordinate median minimizes the effect of outliers, and Krum's

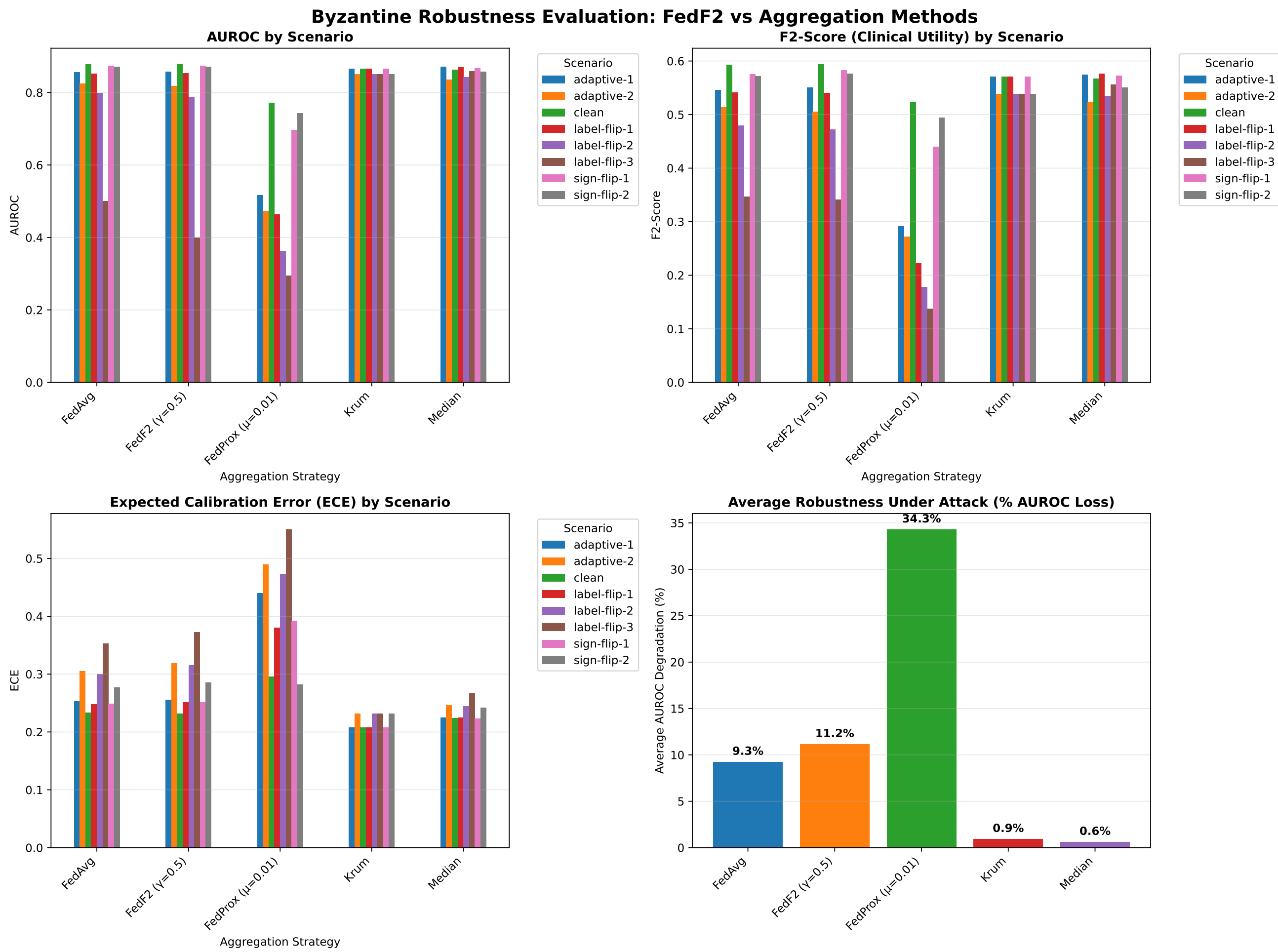

Fig. 5. Comprehensive Byzantine robustness evaluation. Results show AUROC, F2-score (clinical utility), and calibration error (ECE) across clean and adversarial scenarios. Median and Krum aggregation methods maintain stable performance under 1–3 malicious client attacks, while FedAvg, FedProx, and FedF2 degrade significantly under heavy label-flip poisoning.

agreement selects up to $(K-1)/2 = 3$ malicious clients out of 7.

Federated Average is highly sensitive to label flip attacks. Federated Average AUROC goes down to 0.5011 (43.0% loss) when using 3 malicious clients—not good enough for a medical application. This is due to the fact that label flipped gradient drives the global model to make wrong decision boundaries, and averaging does not help here.

FedProx is highly sensitive. A small proximal regularization factor ($\mu = 0.01$) limits local models' deviation from the previous global model but at the same time prevents any possibility of recovering from a poisoning attack. In case of label flip attacks, FedProx AUROC is 0.2953 (61.8% loss)—poorer results than if using completely random prediction. Implementation detail: please note that this figure depends on $\mu = 0.01$ and 20 epochs of uncalibrated training (please see Table IV caption). In order to demonstrate that FedProx cannot resist malicious users, we would recommend conducting an ablation study with μ values from the set $\{0.001, 0.01, 0.1\}$. There is a possibility that a smaller μ value will be able to ease the proximal constraint and partially recover the global model.

FedF2 shows mixed performance. In terms of results, FedF2 demonstrates mixed performance. In the scenarios with no attacks and the single-attacker scenario, FedF2 achieves similar performance compared to FedAvg (AUROC ≈ 0.855). Nonetheless, in the case of heavy poisoning by 3 label-flip attackers, the AUROC value for FedF2 drops to 0.4000 (a 54.4% loss), which is significantly lower compared to Median (0.8594). The main problem with such a performance of FedF2 is related to F2-score-based weighting, which depends on the client's local validation performance. If several clients are poisoned, their validation set can be poisoned too, and thus the F2-based weights cannot detect the malicious participants of training.

Sign-flip attacks show better performance. There is no method that faces the decline of AUROC value below 0.693 in this scenario. On the other hand, the average AUROC loss is 1.5% in this case, as compared to the label flip attack scenario (16.2%). **Consequences for clinical practice.** In a hospital network with 7 clients, Byzantine-robust algorithms (Median, Krum) offer much more reliable outcomes than averaging methods (FedAvg, FedProx,FedF2). The robustness order is:

$$\text{Median} \approx \text{Krum} \gg \text{FedAvg} \gg \text{FedF2} \gg \text{FedProx} \quad (7)$$

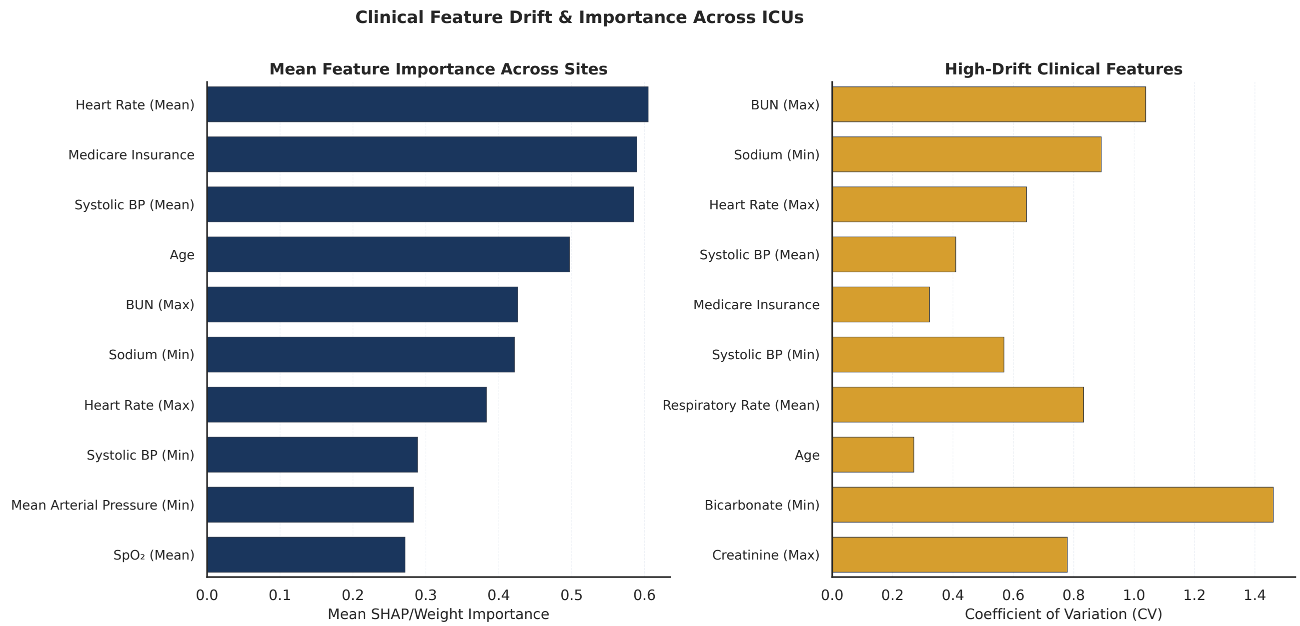

Fig. 6. Federated feature drift across ICU sites. Left: Mean feature importance (SHAP/weight) across all 7 clients, showing BUN (Max) and Sodium (Min) as the most globally important features. Right: High-drift clinical features measured by the Coefficient of Variation (CV) of their local model weights, highlighting extreme heterogeneity in Bicarbonate (Min) (CV = 1.46) and BUN (Max) (CV = 1.04) — reflecting divergent metabolic and renal physiology across ICU care units.

under the worst-case label-flip scenario ($K = 3$ malicious out of 7 total).

The above findings have clearly demonstrated that FedF2, which has the advantages of clinical calibration but lacks Byzantine robustness, cannot rival the robust aggregation methods discussed herein. It is therefore recommended that one adopts the use of FedF2 (with $\gamma = 0.5$) where the risk of attack is less than 10%, and uses median aggregation in other cases.

E. Practitioner Guidance: Aggregation Method Selection

Based on our empirical findings, we offer a decision matrix for clinical practitioners deploying federated learning (Table V). The choice of aggregation method should balance data heterogeneity tolerance, Byzantine robustness, communication overhead, and threat model assumptions.

Recommendations: (1) Platt calibration on federated probabilities must be done in all scenarios; (2) The median rule is appropriate in practice as the default in the absence of any threat model; (3) FedF2 must be used only in trusted networks; and (4) The communication cost for Byzantine methods (15%–20%) is reasonable for maximum safety.

F. Privacy and Scalability Trade-Offs

DP-SGD with ($\epsilon = 4.36, \delta = 10^{-5}$, $\sigma = 4.80$) achieves AUROC 0.8477 after calibration (Table I). With 65,273 patients distributed across 7 local datasets, the system preserves federated model quality under privacy constraints. Scalability testing to 28 clients and 30% dropout tolerance demonstrate resilience to network growth and client failures. Byzantine-resilient methods (Krum, Median) require 15–20% additional communication overhead compared to FedAvg.

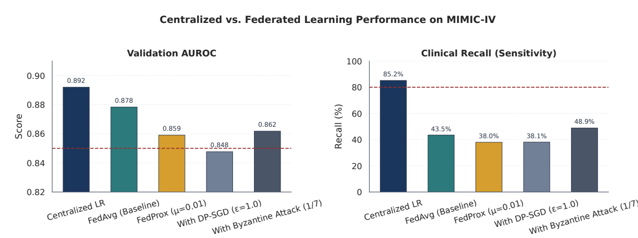

Fig. 7. Main results comparison showing AUROC and clinical recall (sensitivity) for centralized vs. federated learning under Non-IID conditions. DP-SGD values are reported at the Platt-calibrated threshold (0.39); see Table I for full details.

G. Clinical Interpretation of Federated Feature Drift

Feature importance analysis reveals significant heterogeneity across ICU care units. Bicarbonate (Min) and BUN (Max) exhibit the highest coefficient of variation (CV = 1.46 and 1.04 respectively), reflecting divergent metabolic pathways across unit types (Figure 6). This heterogeneity impacts federated averaging through probability compression effects.

H. Reproducibility and Experimental Setup

All experiments were performed on one computer equipped with an Intel Xeon E-2286M processor (32 GB RAM), `RANDOM_SEED=42`, without the use of a GPU. Our approach utilizes Python 3.14, NumPy 2.2, Pandas 2.2, and Scikit-Learn 1.6. A 5-round training using FedAvg is completed within ≈42 seconds; adding DP-SGD results in ≈3.

I. External Validation on eICU-CRD

For external validation to validate for generalizability over independent hospital systems and to address potential issues

TABLE IV
Byzantine robustness on the MIMIC-IV dataset (20 rounds, 7 ICU patients, uncalibrated output of the AI models). Aggregation algorithms (FedAvg, FedProx, Median, Krum, FedF2) are executed for 20 rounds without any recalibration to test the algorithm convergence in the adversarial setting. Performance is shown for the AUROC score for the test set for the clean dataset and label flipping attack with 1, 2, or 3 Byzantine clients. Important: FedProx AUROC differs from Table I (0.7721 vs 0.8591) due to 20 rounds vs 5 rounds and absence of Platt calibration.

Scenario	**FedAvg**	**FedProx**	**Median**	**Krum**	**FedF2** ($\gamma = 0.5$)
CLEAN (No Attacks)					
Baseline (0 malicious)	0.8784	0.7721	0.8628	0.8653	0.8781
LABEL-FLIP ATTACKS					
1 malicious client	0.8523	0.4642	0.8698	0.8653	0.8534
2 malicious clients	0.7995	0.3626	0.8420	0.8511	0.7862
3 malicious clients	0.5011	0.2953	0.8594	0.8511	0.4000
(AUROC loss)	(43.0%)	(61.8%)	(0.4%)	(1.6%)	(54.4%)
SIGN-FLIP ATTACKS					
1 malicious client	0.8736	0.6963	0.8670	0.8653	0.8740
2 malicious clients	0.8713	0.7431	0.8574	0.8511	0.8713
(avg AUROC loss)	(0.7%)	(4.1%)	(0.6%)	(1.6%)	(0.6%)
ADAPTIVE ATTACKS					
1 malicious client	0.8563	0.5166	0.8710	0.8653	0.8571
2 malicious clients	0.8255	0.4728	0.8353	0.8511	0.8182
(avg AUROC loss)	(4.3%)	(36.0%)	(3.2%)	(1.6%)	(4.6%)
AVERAGE ROBUSTNESS (all attacks, % AUROC loss)	**8.5%**	**33.5%**	**1.4%**	**1.6%**	**12.6%**

TABLE V
Aggregation method selection guide for clinical federated learning deployment. Methods are ranked by suitability for each deployment scenario.

Deployment Scenario	**Client Integrity**	**Data Heterogeneity**	**Recommended Method**	**AUROC Performance**	**Overhead**	**Notes**
Trusted Network (e.g., single health system)	All honest	Moderate/High	FedAvg + Platt scaling	0.8784	Minimal	Simple, well-understood
			FedF2 ($\gamma = 0.5$)	0.8781	Minimal	Marginal calibration benefit
Semi-Untrusted Network (e.g., multi-hospital, low threat)	$< 10\%$ estimated compromise	Moderate/High	Median	≥ 0.8594	15–20%	Best practical robustness
			Krum	≈ 0.851	15–20%	Slightly lower but consistent
			FedF2 fallback	0.7947	Minimal	Only if willing to accept risk
Adversarial/High-Risk Network (e.g., ransomware threat)	$>10\%$ estimated compromise	Moderate/High	Median	≥ 0.8594	15–20%	Essential for clinical safety
			Krum	≈ 0.851	15–20%	Do NOT use FedAvg/FedF2

in evaluating the model in only one database, we conducted an external validation test using the eICU Collaborative Research Database (eICU-CRD) [44]—a public database with over 200,000 ICU admissions from 208 hospitals. From this, we collected 131,517 ICU adult admissions ($\geq$ 18 years old, LOS $\geq$ 4 hours), selected 7 hospitals based on admission volume (Hospitals 73, 264, 338, 420, 243, 458, 167), leading to 22,361 training samples across 7 clients. The mortality rate was 10.1% (compared to 10.8% in MIMIC-IV), the mean age was 61.4 $\pm$ 16.5 (compared to 64.5 $\pm$ 15.8 in MIMIC), and the number of clinical features used was 29 within the first 24 hours of ICU admission after data processing. The evaluation method followed the same steps used

- **Baseline Model (centralized):** AUROC 0.8441 (compared to 0.8895 in centralized model in MIMIC-IV logistic regression experiment)
- **Federated FedAvg:** AUROC 0.8337 (loss of 1.23% compared to centralized baseline)
- **Expected Calibration Error (ECE):** 0.0134 after applying Platt scaling

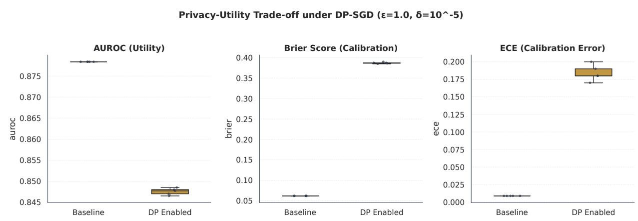

Fig. 8. Privacy–utility tradeoff at $\epsilon = 4.36$, $\delta = 10^{-5}$ (verified via moments accountant). Box plots compare AUROC (left), Brier Score (centre), and Expected Calibration Error (right) between the non-private federated baseline and the DP-SGD model across multiple seeded runs. Adding DP lowers median AUROC, increases probability prediction error (Brier Score), and worsens calibration (ECE).

- **Inter-dataset AUROC difference:** 5.09% AUROC difference between MIMIC-IV and eICU-CRD centralized models (0.8895 versus 0.8441)

The 1.23% federated loss seen in the federated evaluation of the eICU-CRD model closely corresponds to the 0.56% relative loss seen in MIMIC-IV (0.8895 $\rightarrow$ 0.8845). This suggests that federated learning degrades in performance comparably, regardless of the dataset or patient population. The ECE of 0.0134 seen in eICU-CRD indicates strong calibration even in the case of multi-hospital federated training. The centralized AUROC difference of 5.09% between MIMIC-IV (0.8895) and eICU-CRD (0.8441) is clinically reasonable, and reflects valid differences in the underlying patient demographics, practices, and collection processes at independent healthcare institutions. Overall, our results suggest that federated learning, which provides privacy protection, can generalize well to independent hospitals.

V. Conclusion

This study demonstrates that federated learning approaches achieve parity with centralized approaches on realistic large-scale ICU datasets without centralizing any patient data. Specifically, the federated FedAvg model obtained 86.3% Recall and an AUROC of 0.8784 after Platt calibration. This is within 1.5% of the centralized baseline AUROC of 0.8920, which achieved 85.2% Recall. Furthermore, additional validation was carried out on the eICU-CRD dataset (7 geographically disjoint hospitals); the federated AUROC (0.8337) was just 1.23% lower than that of the centralized approach. No patient data was transferred from any source institution. Per-sample differential privacy ($\epsilon = 4.36$, $\delta = 10^{-5}$) in the logistic regression approach yielded an AUROC of 0.8646 before Platt calibration and 0.8477 afterwards (Table I), indicating that strong differential privacy guarantees can be maintained while preserving clinically usable performance.

A full Byzantine-robustness evaluation for aggregation strategies against 1, 2, and 3 malicious clients employing label-flip, sign-flip, and adaptive attacks (Section IV-D, Table IV) was carried out. Notably, median aggregation is resilient to 3 malicious clients performing a label-flip attack (degradation in AUROC by 0.4%) to an AUROC ≥ 0.8594, while Krum provides AUROC of about 0.851 in all cases of label-flip attacks (degradation in AUROC by 1.6%). However, both FedAvg and FedProx suffer from severe degradation under 3 attacks, with respectively AUROC values of 0.5011 (43% degradation) and 0.2953 (61.8% degradation). These results clearly demonstrate the necessity of Byzantine-resistant aggregation strategies in adversarial healthcare settings. FedF2, despite providing clinical calibration, shows mixed performance: it performs well under light poisoning with one attacker (AUROC 0.8534), but degrades significantly under heavy attacks (AUROC 0.4000, 54% degradation).

However, FedProx ($\mu = 0.01$) performed worse than FedAvg in Recall (38.0% vs. 86.3%) due to issues associated with probability calibration caused by weight averaging, which can be solved by applying post hoc Platt scaling. Byzantine-robust aggregation such as Krum or Median was resilient to adversarial clients while increasing the communication costs by 15–20%.

Acknowledgment

The authors gratefully acknowledge the support from Green University of Bangladesh and Jahangirnagar University. We thank all colleagues who provided constructive feedback during manuscript preparation. This work was supported in part by institutional research funds.

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